

Point-By-Point Response for Reviewer 2

(with Reviewer's text in black, response in blue, corresponding actions taken in bold marked by "**Action:**")

I like to thank the authors for addressing all main issues during the revision of the manuscript. I just have some minor points, that need to be resolved until final publication:

Many thanks from us back to you for your thoroughness and guidance! Your highly useful suggestions led to a much better paper and sparked some important optimizations and improvements of our system.

Page 5: Here the authors use the abbreviation FPP for the first time. Please write the full words on first occurrence of an abbreviation.

Action: Done.

Page 13: "...fastq files from Table 2 well as for simulated fastq..." => "...fastq files from Table 2 as well as for simulated fastq...". "as" is missing

Action: Done.

Page 13: "... database against the corresponding complete viral database and under KrakenUniq ...". Remove "and"

Action: Done.

Page 16: "... during the update phase for the sake for contaminant detection." Exchange "for" with "of"

Action: Done.

Page 16: "... long reads based on a set genomes and ..." => "... long reads based on a set of genomes and ..."

Action: Done.

Supplements:

Page3: The author should consider defining the range for r.

Action: Ok – we added a corresponding note.

Page 3: "two skip-1-repeat lengths for $s = \text{ACTCT}$ are $\text{srl}_1(s; 0) = 0$ and $\text{srl}_1(s; 1) = 2$ " Here, either the two examples or the range definition of i above is incorrect. Because from definition above $1 \leq i \leq |s|$, we know that i cannot be 0.

Action: The two examples were flawed indeed. We corrected them as follows: " $\text{srl}_1(s, 1) = 0$ and $\text{srl}_1(s, 2) = 2$ ". (We also fixed a typo.)

Page 3: "e.g. $\text{sr}_0(\text{AAAA}; 1) = 3$ but also $\text{sr}_1(\text{AAAA}; 1) = 1$." Both should be sr_0 and sr_1

Action: Done.